

Physics 514, Problem Set 2: Tensors and Coordinate Transformations

due: *Tuesday, February 1 at Noon*

Please place your solutions in the box outside my office (Rutherford 316) before the due date. You are encouraged to discuss these problems with your colleagues, but you must write up your own solutions; the solutions you hand in should reflect your own work and understanding.

Reading: Chapter 2 of Carroll.

1. Chapter 2, Problem 6.
2. Chapter 2, Problem 7
3. Chapter 2, Problem 8.
4. Consider a D dimensional manifold. In any coordinate system x^μ , the Kronecker delta symbol δ^μ_ν is defined to be equal to 1 if $\mu = \nu$ and zero otherwise. Prove that δ^μ_ν is a tensor of weight (1,1).
5. One way of thinking of a vector field is as an infinitesimal coordinate transformation. Consider a one parameter family of coordinate transformations $x^\mu \rightarrow y^\mu = y^\mu(x^\mu, \lambda)$ where λ is a continuous parameter. We will take the coordinate transformation to be trivial at $\lambda = 0$, i.e. $y^\mu(x^\mu, 0) = x^\mu$, and define

$$v^\mu = \frac{\partial y^\mu}{\partial \lambda}$$

Since the limit of small λ

$$y^\mu = x^\mu + \lambda v^\mu + \dots$$

we sometimes say that the vector v^μ "generates" this infinitesimal coordinate transformation.

- (a) Consider three dimensional flat space in cartesian coordinates x^i , $i = 1, 2, 3$. Show that the vectors

$$\partial_1, \quad \partial_2, \quad \partial_3$$

generate infinitesimal translations in the x^1 , x^2 , and x^3 directions, respectively. Note that we are using here the notation where a vector v^μ is written in "index free" notation as $v = v^\mu \partial_\mu$. So, for example ∂_1 is the vector which has components (1,0,0) in Cartesian coordinates.

- (b) In three dimensional flat space, define three vectors v, w, z by

$$v = x^2\partial_3 - x^3\partial_2, \quad w = x^3\partial_1 - x^1\partial_3, \quad z = x^1\partial_2 - x^2\partial_1$$

Show that these vectors generate infinitesimal rotations around the x^1 , x^2 , and x^3 axes, respectively.

6. For two vectors v^μ and w^μ we define the commutator, or Lie bracket, of the vectors to be

$$[v, w]^\nu = v^\mu \partial_\mu w^\nu - w^\mu \partial_\mu v^\nu$$

- (a) Show that $[v, w]^\nu$ transforms like a vector under coordinate transformations.
- (b) Consider the three vectors v, w, z defined above. Compute the commutators between these vectors. These commutators form what is known as the $SU(2)$ or $SO(3)$ algebra, which you probably encountered in your quantum mechanics class as the algebra of angular momenta.

Problem 1

Carroll 2.6

6. Consider $\mathbf{R}^3$ as a manifold with the flat Euclidean metric, and coordinates $\{x, y, z\}$. Introduce spherical polar coordinates $\{r, \theta, \phi\}$ related to $\{x, y, z\}$ by

$$\begin{aligned}x &= r \sin \theta \cos \phi \\y &= r \sin \theta \sin \phi \\z &= r \cos \theta,\end{aligned}\tag{2.99}$$

so that the metric takes the form

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.\tag{2.100}$$

- (a) A particle moves along a parameterized curve given by

$$x(\lambda) = \cos \lambda, \quad y(\lambda) = \sin \lambda, \quad z(\lambda) = \lambda.\tag{2.101}$$

Express the path of the curve in the $\{r, \theta, \phi\}$ system.

- (b) Calculate the components of the tangent vector to the curve in both the Cartesian and spherical polar coordinate systems.

$$a. \quad r = (x^2 + y^2 + z^2)^{1/2} = \sqrt{1 + \lambda^2}$$

$$\theta = \arctan\left(\frac{\sqrt{x^2 + y^2}}{z}\right) = \arctan\left(\frac{1}{\lambda}\right)$$

$$\phi = \arctan\left(\frac{y}{x}\right) = \lambda.$$

b. Cartesian:

$$\frac{dx}{d\lambda} = -\sin \lambda$$

$$\frac{dy}{d\lambda} = \cos \lambda$$

$$\frac{dz}{d\lambda} = 1$$

Spherical:

$$\frac{dr}{d\lambda} = \frac{\lambda}{\sqrt{1+\lambda^2}}$$

$$\frac{d\theta}{d\lambda} = -\frac{1}{1+\lambda^2}$$

$$\frac{d\phi}{d\lambda} = 1$$

Problem 2

Carroll 2-7

7. Prolate spheroidal coordinates can be used to simplify the Kepler problem in celestial mechanics. They are related to the usual cartesian coordinates (x, y, z) of Euclidean three-space by

$$x = \sinh \chi \sin \theta \cos \phi,$$

$$y = \sinh \chi \sin \theta \sin \phi,$$

$$z = \cosh \chi \cos \theta.$$

Restrict your attention to the plane $y = 0$ and answer the following questions.

(a) What is the coordinate transformation matrix $\partial x^\mu / \partial x^{\nu'}$ relating (x, z) to (χ, θ) ?

(b) What does the line element ds^2 look like in prolate spheroidal coordinates?

$$a. \begin{pmatrix} \cosh \chi \sin \theta \cos \phi & \sinh \chi \cos \theta \cos \phi \\ \sinh \chi \cos \theta & -\cosh \chi \sin \theta \end{pmatrix}$$

$$b. \begin{aligned} dx &= \cosh \chi \sin \theta \cos \phi d\chi \\ &\quad + \sinh \chi \cos \theta \cos \phi d\theta \\ &\quad - \sinh \chi \sin \theta \sin \phi d\phi \end{aligned}$$

$$\begin{aligned} dy &= \cosh \chi \sin \theta \sin \phi d\chi \\ &\quad + \sinh \chi \cos \theta \sin \phi d\theta \\ &\quad + \sinh \chi \sin \theta \cos \phi d\phi \end{aligned}$$

$$dz = \sinh \chi \cos \theta d\chi - \cosh \chi \sin \theta d\theta$$

Ugly week:

$$\begin{aligned}
 (dx)^2 &= \cosh^2 X \sin^2 \theta \cos^2 \phi (dX)^2 \\
 &\quad + 2 \cosh X \sinh X \sin \theta \cos \theta \cos^2 \phi dX d\theta \\
 &\quad - 2 \cosh X \sinh X \sin^2 \theta \cos \phi \sin \phi dX d\phi \\
 &\quad + \sinh^2 X \cos^2 \theta \cos^2 \phi (d\theta)^2 \\
 &\quad - 2 \sinh^2 X \cos \theta \sin \theta \cos \phi \sin \phi d\theta d\phi \\
 &\quad + \sinh^2 X \sin^2 \theta \sin^2 \phi (d\phi)^2
 \end{aligned}$$

$$\begin{aligned}
 (dy)^2 &= \cosh^2 X \sin^2 \theta \sin^2 \phi (dX)^2 \\
 &\quad + 2 \cosh X \sinh X \sin \theta \cos \theta \sin^2 \phi dX d\theta \\
 &\quad + 2 \cosh X \sinh X \sin^2 \theta \cos \phi \sin \phi dX d\phi \\
 &\quad + \sinh^2 X \cos^2 \theta \sin^2 \phi (d\theta)^2 \\
 &\quad + 2 \sinh^2 X \cos \theta \sin \theta \cos \phi \sin \phi d\theta d\phi \\
 &\quad + \sinh^2 X \sin^2 \theta \cos^2 \phi (d\phi)^2
 \end{aligned}$$

$$\begin{aligned}
 (dz)^2 &= \sinh^2 X \cos^2 \theta (dX)^2 \\
 &\quad - 2 \sinh X \cosh X \cos \theta \sin \theta dX d\theta \\
 &\quad + \cosh^2 X \sin^2 \theta (d\theta)^2
 \end{aligned}$$

$$\begin{aligned}
 (dx)^2 + (dy)^2 = & \cosh^2 X \sin^2 \Theta (dX)^2 \\
 & + 2 \cosh X \sinh X \cos \Theta \sin \Theta dX d\Theta \\
 & + \sinh^2 X \cos^2 \Theta (d\Theta)^2 \\
 & + \sinh^2 X \sin^2 \Theta (d\phi)^2
 \end{aligned}$$

Finally, $ds^2 = dx^2 + dy^2 + dz^2$

$$\begin{aligned}
 = & (dX)^2 (\sinh^2 X \cos^2 \Theta + \cosh^2 X \sin^2 \Theta) \\
 & + (d\Theta)^2 (\cosh^2 X \sin^2 \Theta + \sinh^2 X \cos^2 \Theta) \\
 & + (d\phi)^2 (\sinh^2 X \sin^2 \Theta) .
 \end{aligned}$$

$$\begin{aligned}
 = & (dX)^2 (\sinh^2 X + \sin^2 \Theta) \\
 & + (d\Theta)^2 (\sinh^2 X + \sin^2 \Theta) \\
 & + (d\phi)^2 (\sinh^2 X \sin^2 \Theta) .
 \end{aligned}$$

Problem 3

Carroll 2-8.

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge (d\eta).$$

$$(A \wedge B)_{\mu_1 \dots \mu_{p+q}} = \frac{(p+q)!}{p! q!} A_{[\mu_1 \dots \mu_p} B_{\mu_{p+1} \dots \mu_{p+q}]}.$$

$$(dA)_{\mu_1 \dots \mu_{p+1}} = (p+1) \partial_{[\mu_1} A_{\mu_2 \dots \mu_{p+1}]}.$$

$$\begin{aligned} d(\omega \wedge \eta)_{\mu_1 \dots \mu_{p+q+1}} &= (p+q+1) \partial_{[\mu_1} (\omega \wedge \eta)_{\mu_2 \dots \mu_{p+q+1}]} \\ &= \frac{(p+q+1)!}{p! q!} \partial_{[\mu_1} \left(\omega_{[\mu_2 \dots \mu_{p+1}} \eta_{\mu_{p+2} \dots \mu_{p+q+1}]} \right) \end{aligned}$$

The inner square brackets drop.

Here is an outline to prove the general case:

$$\begin{aligned} &A_{[\mu_1 \dots \mu_p} B_{[\mu_{p+1} \dots \mu_{p+q}} C_{\mu_{p+q+1} \dots \mu_{p+q+r}]} \\ &= A_{[\mu_1 \dots \mu_p} B_{\mu_{p+1} \dots \mu_{p+q}} C_{\mu_{p+q+1} \dots \mu_{p+q+r}]} \end{aligned}$$

First, expand both sides.

Group together all terms with the same indices on A .

Each group's elements on the LHS has $(q+r)!$ terms which are all identical to each other, so each group sums up to:

$$(q+r)! A_{v_1 \dots v_p} B_{[v_{p+1} \dots v_{p+q}} C_{v_{p+q+1} \dots v_{p+q+r}]}$$

$$\text{where } v_i \rightarrow \mu_j.$$

By definition of antisymmetrization this is:

$$A_{v_1 \dots v_p} (\text{alternating sum over } B, C)$$

But this is exactly the corresponding group on the RHS.

Thus, $LHS = RHS$.

Continuing on, then,

$$\begin{aligned} & d(\omega \wedge \eta)_{\mu_1 \dots \mu_{p+q+1}} \\ &= \frac{(p+q+1)!}{p!q!} \partial_{[\mu_1} (\omega_{\mu_2 \dots \mu_{p+1}}) \eta_{\mu_{p+2} \dots \mu_{p+q+1}}] \\ & \quad + \frac{(p+q+1)!}{p!q!} \omega_{[\mu_2 \dots \mu_{p+1}} \partial_{\mu_1} \eta_{\mu_{p+2} \dots \mu_{p+q+1}}] \end{aligned}$$

& you can pretty easily show the
rest.

Problem 4

4. Consider a D dimensional manifold. In any coordinate system x^μ , the Kronecker delta symbol δ^μ_ν is defined to be equal to 1 if $\mu = \nu$ and zero otherwise. Prove that δ^μ_ν is a tensor of weight (1, 1).

By definition:
$$x^{\mu'} = x^{\nu'} g^{\mu'}_{\nu'}$$
$$= \frac{\partial x^{\nu'}}{\partial x^{\nu}} x^{\nu} g^{\mu'}_{\nu'}$$

Multiplying both sides by $\frac{\partial x^{\mu}}{\partial x^{\mu'}}$,

$$x^{\mu} = \left(\frac{\partial x^{\nu'}}{\partial x^{\nu}} \frac{\partial x^{\mu}}{\partial x^{\mu'}} g^{\mu'}_{\nu'} \right) x^{\nu}$$

So that:
$$g^{\mu}_{\nu} = \frac{\partial x^{\nu'}}{\partial x^{\nu}} \frac{\partial x^{\mu}}{\partial x^{\mu'}} g^{\mu'}_{\nu'}$$

$\Rightarrow g^{\mu}_{\nu}$ transforms like
a tensor.

Problem 5

5. One way of thinking of a vector field is as an infinitesimal coordinate transformation. Consider a one parameter family of coordinate transformations $x^\mu \rightarrow y^\mu = y^\mu(x^\mu, \lambda)$ where λ is a continuous parameter. We will take the coordinate transformation to be trivial at $\lambda = 0$, i.e. $y^\mu(x^\mu, 0) = x^\mu$, and define

$$v^\mu = \frac{\partial y^\mu}{\partial \lambda}$$

Since the limit of small λ

$$y^\mu = x^\mu + \lambda v^\mu + \dots$$

we sometimes say that the vector v^μ "generates" this infinitesimal coordinate transformation.

- (a) Consider three dimensional flat space in cartesian coordinates x^i , $i = 1, 2, 3$. Show that the vectors

$$\partial_1, \quad \partial_2, \quad \partial_3$$

generate infinitesimal translations in the x^1 , x^2 , and x^3 directions, respectively. Note that we are using here the notation where a vector v^μ is written in "index free" notation as $v = v^\mu \partial_\mu$. So, for example ∂_1 is the vector which has components $(1, 0, 0)$ in Cartesian coordinates.

Letting $v = \partial_1$, we see that:

$$\frac{\partial y^\mu}{\partial \lambda} \partial_\mu = \partial_1$$

So that:

$$\frac{\partial y^1}{\partial \lambda} = 1, \quad \text{others are zero.}$$

$$\Rightarrow y^\mu \partial_\mu = x^\mu \partial_\mu + \lambda \partial_1 \quad \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} x_1 + \lambda \\ x_2 \\ x_3 \end{pmatrix}$$

& the argument is the same for ∂_2, ∂_3 .

(b) In three dimensional flat space, define three vectors v, w, z by

$$v = x^2 \partial_3 - x^3 \partial_2, \quad w = x^3 \partial_1 - x^1 \partial_3, \quad z = x^1 \partial_2 - x^2 \partial_1$$

Show that these vectors generate infinitesimal rotations around the x^1 , x^2 , and x^3 axes, respectively.

For v :

$$\frac{\partial y^\mu}{\partial \lambda} \partial_\mu = x^2 \partial_3 - x^3 \partial_2$$

$$\Rightarrow \frac{\partial y^3}{\partial \lambda} = x^2, \quad \frac{\partial y^2}{\partial \lambda} = -x^3, \quad \frac{\partial y^1}{\partial \lambda} = 0.$$

$$\Rightarrow y^\mu \partial_\mu = x^\mu \partial_\mu + \lambda x^2 \partial_3 - \lambda x^3 \partial_2$$

$$\begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix} = \begin{pmatrix} x^1 \\ x^2 + \lambda x^3 \\ x^3 - \lambda x^2 \end{pmatrix}$$

Suppose we have a circle around the x^1 axis, parametrized as $r(\cos\theta, \sin\theta)$

$$\begin{aligned}
\text{Then } r \begin{pmatrix} \cos(\theta + \lambda) \\ \sin(\theta + \lambda) \end{pmatrix} &= r \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \\
&= r \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \lambda \quad (\text{for small } \lambda) \\
&= \begin{pmatrix} -x_2 \\ x_3 \end{pmatrix} \lambda \quad \text{for a rotation} \\
&\quad \text{around } x^1.
\end{aligned}$$

This is for a counterclockwise rotation,
so v generates a clockwise rotation.

w, z are similar.

Problem 6

6. For two vectors v^μ and w^μ we define the commutator, or Lie bracket, of the vectors to be

$$[v, w]^\nu = v^\mu \partial_\mu w^\nu - w^\mu \partial_\mu v^\nu$$

- (a) Show that $[v, w]^\nu$ transforms like a vector under coordinate transformations.
- (b) Consider the three vectors v, w, z defined above. Compute the commutators between these vectors. These commutators form what is known as the $SU(2)$ or $SO(3)$ algebra, which you probably encountered in your quantum mechanics class as the algebra of angular momenta.

$$a. [v, w]^\nu = v^\mu \partial_\mu w^\nu - w^\mu \partial_\mu v^\nu$$

$$= v^{\mu'} \partial_{\mu'} \left(\frac{\partial x^\nu}{\partial x^{\nu'}} w^{\nu'} \right) - w^{\mu'} \partial_{\mu'} \left(\frac{\partial x^\nu}{\partial x^{\nu'}} v^{\nu'} \right)$$

$$= v^{\mu'} w^{\nu'} \partial_{\mu'} \left(\frac{\partial x^\nu}{\partial x^{\nu'}} \right) - w^{\mu'} v^{\nu'} \partial_{\mu'} \left(\frac{\partial x^\nu}{\partial x^{\nu'}} \right)$$

$$+ \frac{\partial x^{\nu'}}{\partial x^\nu} (v^{\mu'} \partial_{\mu'} w^{\nu'} - w^{\mu'} \partial_{\mu'} v^{\nu'})$$

$$= v^{\mu'} w^{\nu'} \partial_{\nu'} \left(\frac{\partial x^\nu}{\partial x^{\mu'}} \right) - w^{\mu'} v^{\nu'} \partial_{\mu'} \left(\frac{\partial x^\nu}{\partial x^{\nu'}} \right)$$

$$+ \frac{\partial x^{\nu'}}{\partial x^\nu} [v, w]^\nu$$

$$= \frac{\partial x^{v'}}{\partial x^v} [v, w]^{v'} \quad \text{as required.}$$

b.

$$v = x^2 \partial_3 - x^3 \partial_2, \quad w = x^3 \partial_1 - x^1 \partial_3, \quad z = x^1 \partial_2 - x^2 \partial_1$$

$$\begin{aligned} [v, w]^{v'} \partial_{v'} &= x^2 \partial_3 (x^3 \partial_1) \\ &\quad + x^2 \partial_3 (-x^1 \partial_3) \\ &\quad - x^3 \partial_2 (x^3 \partial_1) \\ &\quad - x^3 \partial_2 (-x^1 \partial_3) \\ &\quad - x^3 \partial_1 (x^2 \partial_3) \\ &\quad - x^3 \partial_1 (-x^3 \partial_2) \\ &\quad + x^1 \partial_3 (x^2 \partial_3) \\ &\quad + x^1 \partial_3 (-x^3 \partial_2) \end{aligned}$$

$$= x^2 \partial_1 - x^1 \partial_2 = -z$$

Cycle the indices to get the others.